

RBE 4540/595 – Homework 3: Grasp Quality Measures

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1 Derivation of the Grasp Matrix

1.1 Object and Contact Geometry

The object is a $6\text{ cm} \times 3\text{ cm}$ rectangle. The origin is placed at the object's geometric center. The contacts, four in number, with inward unit normals, are:

Contact	Position p_i (cm)	Inward normal $\hat{n}_i$
1 (top-left)	$(-1, 1.5)$	$(0, -1)$
2 (top-right)	$(1, 1.5)$	$(0, -1)$
3 (left)	$(-3, 0.5)$	$(1, 0)$
4 (bottom)	$(0, -1.5)$	$(0, 1)$

1.2 Full Spatial Grasp Matrix

Under the **hard-finger contact model**, each contact transmits force only (no torque) along three directions in its local contact frame: the inward normal $\hat{n}_i$ and two tangential directions $\hat{t}_{i1}, \hat{t}_{i2}$. Let $R_i = [\hat{n}_i \ \hat{t}_{i1} \ \hat{t}_{i2}] \in \mathbb{R}^{3 \times 3}$ be the rotation from the contact frame to the object frame, and let $p_i \in \mathbb{R}^3$ be the contact position relative to the object center. The skew matrix of the moment-arm is:

$$[p_i]_{\times} = \begin{bmatrix} 0 & -p_{i,z} & p_{i,y} \\ p_{i,z} & 0 & -p_{i,x} \\ -p_{i,y} & p_{i,x} & 0 \end{bmatrix}.$$

Each contact yields a 6×3 block

$$G_i = \begin{bmatrix} R_i \\ [p_i]_{\times} R_i \end{bmatrix} \in \mathbb{R}^{6 \times 3},$$

mapping the local contact force $f_i \in \mathbb{R}^3$ to the object wrench $w_i = G_i f_i \in \mathbb{R}^6$, ordered $(F_x, F_y, F_z, \tau_x, \tau_y, \tau_z)$. The full spatial grasp matrix for n contacts is the horizontal concatenation

$$G = [G_1 \ G_2 \ \cdots \ G_n] \in \mathbb{R}^{6 \times 3n}.$$

1.3 Reduction to the Planar Grasp Matrix

Because all contact points and force directions lie in the object's xy -plane ($p_{i,z} = 0$ for every contact), the planar wrench components of interest are F_x, F_y, τ_z — rows 1, 2, and 6 of G . The remaining rows, F_z (row 3), τ_x (row 4), and τ_y (row 5), are **eliminated**: they correspond to out-of-plane force and out-of-plane moment components that do not exist in a 2D problem. This can

also be verified directly: with $p_{i,z} = 0$, rows 4 and 5 of $[p_i]_{\times} R_i$ depend only on the out-of-plane force component (third column of R_i), and row 3 of R_i itself is zero for any in-plane contact frame — so rows 3, 4, 5 carry no information relevant to a planar grasp.

The reduced planar grasp matrix is

$$G_p = [G_1^p \quad G_2^p \quad \cdots \quad G_n^p] \in \mathbb{R}^{3 \times 3n}, \quad G_i^p = \begin{bmatrix} n_{x,i} & t_{x,i} & 0 \\ n_{y,i} & t_{y,i} & 0 \\ p_{x,i}n_{y,i} - p_{y,i}n_{x,i} & p_{x,i}t_{y,i} - p_{y,i}t_{x,i} & 0 \end{bmatrix},$$

where $\hat{t}_i = (-n_{y,i}, n_{x,i})$ is the in-plane tangential direction (a 90° rotation of $\hat{n}_i$). The third column of each block, corresponding to the out-of-plane force direction, evaluates to zero identically once rows 3–5 are dropped; it is retained in the matrix below only because the assignment specifies eliminating rows, not columns.

Substituting the four fixed contacts:

$$G_p = \left[\begin{array}{ccc|ccc|ccc|ccc} 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \\ -1 & 0 & 0 & -1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1.5 & 0 & -1 & -1.5 & 0 & -0.5 & -3 & 0 & 0 & -1.5 & 0 \end{array} \right]$$

Adding a 5th contact at location p_5 with normal $\hat{n}_5$ appends one more 3×3 block G_5^p computed the same way as discussed above.

1.4 Grasp Quality Metrics

The three grasp quality measures (Roa and Suarez (2014)) are computed from the singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3$ of G_p :

- **Section 3.1.1: Minimum singular value:** $Q_{\text{MSV}} = \sigma_{\min} = \sigma_3$. Computes the worst-case disturbance the grasp can resist in any wrench direction.
- **Section 3.1.2: Volume of the wrench-space ellipsoid:** $Q_{\text{VOL}} = \sigma_1 \sigma_2 \sigma_3 = \sqrt{\det(G_p G_p^\top)}$. Measures overall coverage of the wrench space.
- **Section 3.1.3: Grasp isotropy index:** $Q_{\text{ISO}} = \sigma_{\min} / \sigma_{\max} = \sigma_3 / \sigma_1$. Measures how uniformly the grasp resists disturbances across all directions, independent of overall magnitude.

Before evaluating any metric, $\text{rank}(G_p)$ is checked. A rank-deficient G_p ($\text{rank} < 3$) means the grasp cannot resist all planar wrench components and the quality value is not meaningful.

1.5 Optimal 5th Contact Location

The 5th contact was swept around the object boundary in 0.1 cm increments (perimeter = 18 cm, 180 samples), recomputing G_p and all three metrics at each location. $\text{rank}(G_p) = 3$ held at every sampled location, so all samples yield a valid quality value. Results are as follows:

Metric	Optimal location (cm)	Value
Q_{MSV}	(3.00, -1.50)	2.234
Q_{VOL}	(3.00, -1.50)	27.019
Q_{ISO}	(0.80, -1.50)	0.470

Q_{MSV} and Q_{VOL} agree on the bottom-right corner $(3, -1.5)$: placing the 5th contact there maximizes the moment arm’s contribution to the weakest wrench direction, and the same placement also increases the other two singular values, so both the worst-case and overall-volume metrics reward it identically. Q_{ISO} instead selects a mid-edge point $(0.8, -1.5)$: because isotropy is a *ratio* $\sigma_{\min}/\sigma_{\max}$ rather than an absolute measure, the corner placement, while strong in the worst direction, also makes that contact disproportionately effective at producing torque, inflating $\sigma_{\max}$ and lowering the ratio. A mid-edge location sacrifices peak strength for more uniform coverage across all three wrench directions, which is what Q_{ISO} rewards. This illustrates the general distinction in the paper between quality measures based on disturbance-resistance strength (3.1.1, 3.1.2) and those based on directional balance (3.1.3): they need not identify the same optimal grasp.

2 Plots

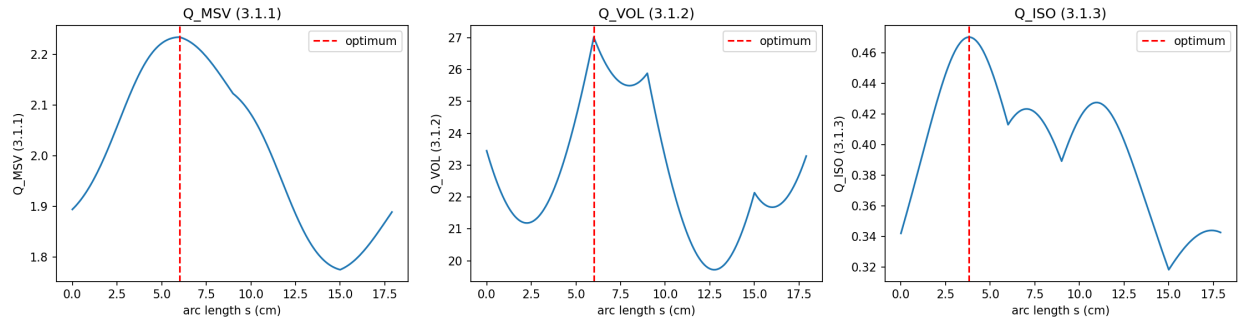


Figure 1: Grasp quality value vs. 5th-contact arc-length position along the object boundary, for each of the three metrics.

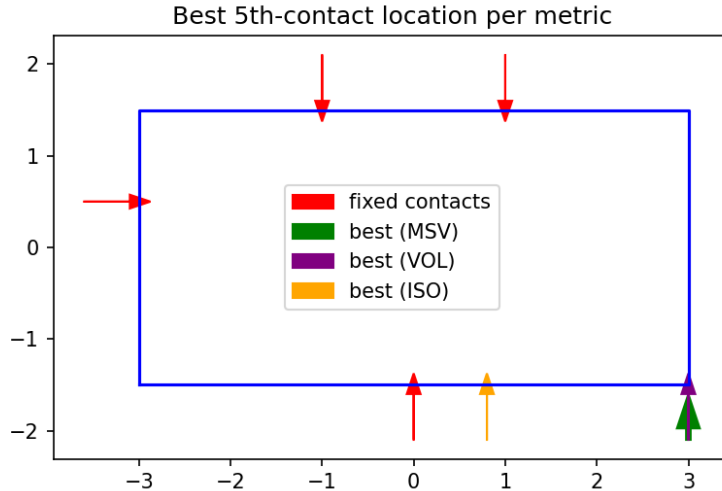


Figure 2: Optimal 5th-contact location for each metric, drawn on the object.